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SIEM SECURITY ANALYSIS REPORT

Generated: 2026-03-26 12:10:00 | Log Source Type: Postfix

Total Events	1000
Parsed Successfully	1000 (100%)
Risk Level	MEDIUM
ML Detected Threats	0

Overall Risk Score: 40/100 (MEDIUM)

1. EXECUTIVE SUMMARY

This comprehensive security report presents the analysis of **1,000** log events from the SIEM system. The analysis successfully parsed **1,000** events (100% success rate) with **0** events failing to parse.

■ **No ML-detected threats** were found in the analyzed log sample. The security posture appears normal.

2. AI-POWERED THREAT ANALYSIS

This section contains detailed threat analysis generated by our AI SOC Analyst.

SIEM Security Analysis Report

Date: March 19, 1900 (Based on log data)

Analyst: AI Security Analyst

1. EXECUTIVE SUMMARY:

The organization's email infrastructure is currently under potential attack, exhibiting signs of brute-force attempts, threat intelligence indicators, and spam activity. While the overall log volume is low and primarily informational, the presence of these alerts necessitates immediate investigation and remediation to prevent further compromise and potential data breaches.

2. KEY FINDINGS:

| Finding | Severity | Description |

|---|---|---|

| Multiple Brute-Force Attempts | HIGH | Two separate alerts indicate brute-force attacks against the mail server. This suggests attackers are attempting to gain unauthorized access to user accounts. |

| Threat Intelligence Match | HIGH | An alert triggered due to a match with threat intelligence feeds. This indicates communication with known malicious IPs or domains, potentially for command and control or data exfiltration. |

| Elevated Spam Activity | MEDIUM | Detection of spam activity suggests the mail server might be compromised or being used to relay spam. This can lead to blacklisting and reputational damage. |

| High Connection Rate from Single IP | MEDIUM | Log entries show a high connection rate (11/3600s) from IP address 45.115.119.123, indicating potential scanning or brute-force activity. |

| Potential Email Spoofing | LOW | The logs show DKIM signatures being added, which is good. However, the presence of spam activity suggests that attackers might be attempting to spoof email addresses to bypass security controls. Further investigation is required to confirm. |

3. ATTACK CHAIN ANALYSIS:

Based on the alerts and log data, the following attack chain can be inferred:

* **Reconnaissance (TA0007):** The high connection rate from IP 45.115.119.123 suggests the attacker is performing network scanning to identify potential targets and vulnerabilities within the email infrastructure.

* **Credential Access (TA0006):** The brute-force alerts indicate attempts to compromise user accounts by guessing passwords. This is a common tactic for gaining initial access.

* **Initial Access (TA0001):** Successful brute-force attempts would grant the attacker initial access to a user account, potentially allowing them to send malicious emails or access sensitive data.

* **Command and Control (TA0011):** The threat intelligence match suggests that a compromised system might be communicating with a command and control server, enabling the attacker to remotely control the system and exfiltrate data.

* **Spam Distribution (TA0015):** The spam activity alert indicates that the mail server might be used to distribute spam emails, potentially damaging the organization's reputation and leading to blacklisting.

4. THREAT INTELLIGENCE:

* **IOCs (Indicators of Compromise):**

* IP Address: 45.115.119.123 (High connection rate, potential scanner)

* IP Addresses: (From threat_intel alert - investigate the specific IPs flagged)

* **Suspicious IPs:**

* 209.85.210.52, 209.85.161.69 (High log count, investigate for malicious activity)

* **Attacker TTPs (Tactics, Techniques, and Procedures):**

* Brute-force attacks (T1110)

* Network scanning (T1046)

* Spam distribution (T1566)

* Potential command and control communication (Investigate based on threat_intel alert)

5. RISK ASSESSMENT:

Current Risk Level: MEDIUM

Justification: The presence of brute-force attempts and threat intelligence matches indicates an active threat targeting the email infrastructure. While the number of error and warning logs is relatively low, the potential impact of a successful attack, including data breaches, reputational damage, and service disruption, warrants a medium-risk assessment. The spam activity further elevates the risk due to potential blacklisting and reputational damage.

****6. RECOMMENDATIONS:****

1. ****Implement Multi-Factor Authentication (MFA):**** Enforce MFA for all email accounts to significantly reduce the risk of successful brute-force attacks. (High Priority)

2. **Investigate and Block Malicious IPs:** Immediately investigate the IPs flagged by the threat intelligence alert and the IP with the high connection rate (45.115.119.123). Block these IPs at the firewall and email gateway to prevent further communication. (High Priority)

3. ****Strengthen Password Policies:**** Enforce strong password policies, including minimum length, complexity requirements, and regular password changes. (High Priority)

4. ****Implement Rate Limiting:**** Configure rate limiting on the mail server to prevent brute-force attacks and excessive connection attempts from single IPs. (Medium Priority)

5. ****Review and Enhance Spam Filtering:**** Review and enhance spam filtering rules to reduce the volume of spam emails and prevent the mail server from being used for spam distribution. Consider implementing SPF, DKIM, and DMARC to prevent email spoofing. (Medium Priority)

6. ****Monitor for Suspicious Activity:**** Implement real-time monitoring and alerting for suspicious activity, such as unusual login attempts, large email volumes, and communication with known malicious IPs. (Medium Priority)

7. ****Conduct Security Awareness Training:**** Educate users about phishing attacks, social engineering, and the importance of strong passwords to reduce the risk of successful attacks. (Low Priority)

****7. INCIDENT RESPONSE:****

1. ****Containment:****

* Immediately block the identified malicious IPs at the firewall and email gateway.

* Isolate any potentially compromised accounts.

2. ****Investigation:****

* Analyze the logs to determine the scope and impact of the attacks.

* Identify any compromised accounts and systems.

*** Investigate the source of the threat intelligence match and the nature of the communication.**

3. ****Eradication:****

* Reset passwords for all compromised accounts.

* Remove any malware or malicious code from infected systems.

- * Clean up any spam queues and remove any traces of spam activity.

4. ****Recovery:****

- * Restore any data that was lost or corrupted.
- * Monitor the systems for any signs of recurrence.

5. ****Post-Incident Activity:****

- * Review and update security policies and procedures.

*** Implement the recommendations outlined in this report.**

- * Conduct a lessons-learned exercise to identify areas for improvement.

This report provides a snapshot of the current security posture based on the provided log data. Continuous monitoring, proactive threat hunting, and regular security assessments are crucial for maintaining a strong security posture and mitigating potential risks.

3. THREAT INTELLIGENCE & IOCs

Top Source IP Addresses (by event count)

Rank	IP Address	Events	Risk Assessment
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

Most Targeted User Accounts

Rank	Username	Events	Priority
1	iqac@tmv.edu.in	18	MEDIUM
2	infocenter@tmv.edu.in	14	MEDIUM
3	finance@tmv.edu.in	11	MEDIUM
4	0.001	10	MEDIUM
5	examination@tmv.edu.in	10	MEDIUM
6	kulguru@tmv.edu.in	8	MEDIUM
7	law@tmv.edu.in	8	MEDIUM
8	administration@tmv.edu.in	8	MEDIUM
9	phd@tmv.edu.in	6	MEDIUM
10	tmvcompscience@gmail.com	6	MEDIUM

5. SECURITY ALERTS

Total alerts generated: 4. Showing top 15 alerts.

Alert #1: bruteforce

Timestamp: 1900-03-18T16:11:13 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

Alert #2: threat_intel

Timestamp: 1900-03-18T16:28:39 | **Source:** N/A

Message: Attack detected: threat_intel (confidence: 80%)

Alert #3: spam_activity

Timestamp: 1900-03-18T16:29:07 | **Source:** N/A

Message: Attack detected: spam_activity (confidence: 60%)

Alert #4: bruteforce

Timestamp: 1900-03-18T16:34:43 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

6. LOG ANALYTICS & STATISTICS

6.1 Severity Distribution

Severity Level	Event Count	Percentage
CRITICAL	0	0.0%
ERROR	6	0.6%
WARNING	7	0.7%
INFO	987	98.7%
DEBUG	0	0.0%

6.2 Log Type Distribution

Log Type	Count
postfix	1000

6.3 Top Source IP Addresses

Rank	IP Address	Event Count	Activity Level
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

7. RECOMMENDATIONS & REMEDIATION

◆ Critical

Immediately investigate all critical and high-severity alerts. Isolate affected systems if compromise is suspected.

◆ High Priority

Review and block suspicious IP addresses with high event counts. Implement additional monitoring on targeted accounts.

◆ Medium Priority

Enable enhanced logging for critical systems. Review access controls for accounts with failed authentication attempts.

◆ Best Practices

Ensure all systems are patched and up-to-date. Implement MFA for all privileged accounts. Regular review of security logs.

◆ Monitoring

Set up real-time alerts for threshold-based detections. Create correlation rules for multi-stage attack patterns.

◆ Response

Document and test incident response procedures. Ensure backup and recovery mechanisms are functional.

8. APPENDIX

Report Metadata

Report Generated	2026-03-26 12:10:00
Log Source Type	Postfix
Total Lines	0
Parsed Lines	0
Failed Lines	0
ML Attacks Detected	0
Total Alerts	4
Analysis Engine	FreeKhana SIEM v2.0 + AI SOC Analyst

Disclaimer: This report is generated automatically by FreeKhana SIEM system. The analysis is based on log data and machine learning models. Human review is recommended for critical security decisions.